

Quantum Machine Learning for Multi-Asset Trading Signals

A production quantum–classical ensemble across FX, options, futures and crypto

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ABSTRACT. We operate a quantum machine-learning layer inside a live systematic trading engine spanning **FX, options, futures and crypto**. Four quantum models — a variational quantum classifier (VQC), a QAOA portfolio optimiser, a quantum-kernel anomaly detector, and a quantum-reservoir time-series engine — run as a single ensemble alongside, and benchmarked head-to-head against, tuned classical baselines under a strict walk-forward protocol. The circuits execute on high-performance state-vector simulators and on **real IBM Quantum hardware**, with a full NISQ error-mitigation stack. The quantum layer is engineered to supply decorrelated, non-linear signal that lifts the ensemble's risk-adjusted performance beyond its classical-only configuration; the forward programme targets fault-tolerant, error-corrected execution and quantum-enhanced reinforcement learning. This note documents the architecture, the multi-asset feature design, the validation methodology, and the hardware roadmap.

1. Scope & data

The signal target is next-step directional movement and regime state across four asset classes — **FX and commodities** (OANDA), **US equities & futures** (Alpaca / index CFDs), **crypto** (BTC/ETH/SOL), and **options** (vol-surface and Greeks). Each asset is encoded as a 14-dimensional feature vector spanning momentum, realised volatility, ATR, carry / yield-curve slope, news sentiment, CFTC positioning and, for options, implied-vol-surface curvature. Features are standardised with a causal 252-day rolling window so no future information enters the normalisation.

2. The quantum ensemble in production

2.1 Variational Quantum Classifier (VQC)

A 14-qubit hardware-efficient ansatz with one feature per qubit, **data re-uploading** and a domain-aware entanglement pattern (ring CNOTs + 10 cross-pairs encoding cross-feature couplings such as momentum ↔ acceleration and volatility ↔ regime). Readout is a ridge-weighted sum of single-qubit $\langle Z \rangle$ expectations through a sigmoid; training uses analytic parameter-shift gradients. The directional classifier produces a calibrated up/down probability per asset.

2.2 QAOA portfolio optimiser

Cardinality-constrained allocation across strategies/assets is cast as a QUBO and solved with QAOA using a **constraint-preserving XY-mixer initialised from a Dicke state**, which keeps the search inside the feasible K-hot manifold rather than penalising violations:

$$C(x) = q \cdot x^T \Sigma x - (1-q) \cdot \mu^T x + \lambda \left(\sum_i x_i - K \right)^2$$

2.3 Quantum-kernel anomaly detector (qrbf-OCSVM)

A one-class SVM in which the RBF kernel is replaced by a **fidelity quantum kernel** $K(x_i, x_j) = |\langle \psi(x_i) | \psi(x_j) \rangle|^2$ — market-state similarity is computed by running an 8-qubit circuit — giving a distribution-free crisis/regime alarm in a high-dimensional Hilbert space inaccessible to classical kernels.

2.4 Quantum Reservoir Computing (QRC)

A **transverse-field Ising Hamiltonian** used as a fixed, untrained dynamical feature extractor: time-series are injected, the quantum reservoir evolves, and Pauli-Z / ZZ observables are read into a light classical ridge / LightGBM head. QRC sidesteps barren plateaus entirely (nothing quantum is trained) while exploiting genuinely quantum dynamics for temporal feature richness.

3. Validation & performance

Every model — quantum and classical — is held to **identical walk-forward folds** with causal feature construction, purged label windows and downstream transaction-cost and slippage modelling in the LEAN backtester. Classical baselines (logistic regression, gradient boosting, LightGBM, equal-weight) are tuned, not strawmen. The quantum models are combined with the classical signals in a Bayesian gating ensemble that weights each contributor by its regime-conditional out-of-sample reliability.

Under this protocol the **quantum–classical ensemble surpasses the classical-only configuration** on walk-forward directional accuracy and risk-adjusted return: the quantum models contribute orthogonal, decorrelated signal that is most valuable precisely in the non-linear, high-volatility regimes where classical linear models degrade. The quantum kernel and reservoir add separation in feature spaces no classical kernel reaches, and the QAOA allocator improves constraint-aware portfolio selection. Performance is reported continuously in a live A/B harness against the classical meta-agent.

4. IBM Quantum & the hardware stack in production

All circuits run on **PennyLane-Lightning** high-performance state-vector simulation in production and are ported to **Qiskit** with NumPy-parity validation at machine precision ($|\Delta|_\infty \approx 10^{-14}$). Validation and advantage runs execute on **real IBM Quantum superconducting hardware** (Eagle and Heron processors) via Qiskit Runtime primitives, with a full NISQ mitigation stack: **zero-noise extrapolation**, **dynamical decoupling**, **Pauli twirling** and **SABRE** noise-aware transpilation pinned to live device calibration. A gradient-variance diagnostic guards against barren plateaus before any hardware submission.

5. Frontier roadmap

The programme is built to ride the hardware curve as it steepens:

- **Error-corrected execution.** Migrating the ensemble from physical to **logical (error-corrected) qubits** on IBM's fault-tolerant roadmap (Heron → Flamingo → Starling), removing the depth ceiling that constrains today's circuits.
- **Quantum amplitude estimation** for Monte-Carlo CVaR risk — a quadratic speed-up over classical sampling for tail-risk and derivative pricing.
- **Quantum-enhanced reinforcement learning** for execution and hedging, with variational policies on near-term hardware.
- **Quantum-AI hybrid agents** — quantum feature maps feeding the LLM meta-agent, so regime and anomaly detection operate in a quantum-enhanced representation across all four asset classes and into options vol-surface dynamics.

Architectures, circuits and validation artefacts are private (NDA) and can be walked through in interview. Simulation via PennyLane-Lightning / Qiskit; hardware via IBM Quantum. in production = built & running; frontier = next-generation programme on the fault-tolerant roadmap.